Swati Aggrawal

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PROFESSIONAL SUMMARY

M.Sc. Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Published IEEE researcher with hands-on experience developing AI and computer vision solutions using Python, PyTorch, TensorFlow, Scikit-learn, FastAPI, Docker, and AWS. Experienced in Vision Transformers, continual learning, deep learning, and deploying scalable ML applications. Passionate about applying AI to solve real-world industrial problems.

SKILLS

- **Programming Languages:** Python, C++, Java, SQL, Bash
- **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Vision Transformers, Continual Learning, Transfer Learning, PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face Transformers, Generative AI
- **Software Engineering:** Data Structures & Algorithms, Object-Oriented Programming (OOP), FastAPI, REST APIs, Git
- **Cloud & DevOps:** Docker, AWS (ECS, Fargate, CloudWatch), Linux
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Investigated continual learning methods for Vision Transformers using PyTorch, adapter tuning, and Elastic Weight Consolidation to reduce catastrophic forgetting across sequential learning tasks.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

EXPERIENCE

Researcher — Vision Lab, Sapienza Università di Roma (Master's Thesis)

Nov 2025 - May 26

- Conducted research under Prof. Marco Raoul Marini on continual learning for computer vision in the Vision Lab.
- Evaluated pre-trained Vision Transformers on sequential image classification benchmarks.
- Designed and implemented adapter-based learning and Elastic Weight Consolidation (EWC) techniques to mitigate catastrophic forgetting.
- Built PyTorch training and evaluation pipelines, analyzing accuracy, forgetting, and computational efficiency.
- Collaborated with researchers to document findings and compare state-of-the-art continual learning approaches.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

PUBLICATION

Machine Learning for Air Pollution – IEEE Conference Publication (2023)

- Evaluated Random Forest, SVM, Decision Tree, and Neural Network models on environmental datasets.
- Performed comparative analysis using accuracy, robustness, and generalization metrics.
- Authored and presented findings through an IEEE peer-reviewed publication.

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PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an end-to-end AI document comparison platform using FastAPI and Sentence Transformers.
- Designed scalable embedding and retrieval pipelines for semantic document comparison.
- Containerized the application with Docker and deployed on AWS ECS/Fargate with CloudWatch monitoring.
- Built REST APIs and optimized inference latency for production deployment.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated defect detection using CNN, SVM, and Random Forest.
- Built preprocessing, training, and evaluation pipelines.
- Compared deep learning and traditional ML approaches.
- Performed model evaluation using accuracy, confusion matrices, and performance analysis.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer-based embeddings, quantization techniques, and text reconstruction methods.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Analyzed trade-offs between compression efficiency and semantic preservation across multiple quantization levels.
- **Technologies:** Python, PyTorch, Transformer Models, NLP, Sentence Embeddings